

CIVENG

TOOLKIT

Structure — Design Calculation

NSCP 2015

ACI 318-14

AISC 360-16

CHECK FAILED

Governing: column C3 combined axial + flexure, utilization 1.07

PROJECT Lot 12 Residence	SHEET S-3D · Rev A	PREPARED BY Engineer, CE
DATE 2026-07-29	ELEMENT Structure — 3D Model Space	CODES NSCP 2015 · ACI 318-14 · AISC 360-16

1 · DESIGN SUMMARY

MEMBERS 148	STOREYS 4	BASE SHEAR 1244 kN
MAX DRIFT 0.0071	CONCRETE 86.4 m³	

Check	Scope / governing	Ratio	Status
Beam flexure	B12 mid-span	0.91	PASS
Column P-M	C3 base	1.07	FAIL
Drift	storey 2, X	0.64	PASS
Footing bearing	F4	—	PASS

2 · PROJECT & DESIGN DATA

Property 1	value 1	Property 8	value 8
Property 2	value 2	Property 9	value 9
Property 3	value 3	Property 10	value 10
Property 4	value 4	Property 11	value 11
Property 5	value 5	Property 12	value 12
Property 6	value 6	Property 13	value 13
Property 7	value 7	Property 14	value 14

3 · MEMBER SCHEDULES

3.1 Beam schedule

Mark	Size	Bars	Ratio
B1	300×550	4ø20	0.50
B2	300×550	4ø20	0.55
B3	300×550	4ø20	0.60
B4	300×550	4ø20	0.65
B5	300×550	4ø20	0.70
B6	300×550	4ø20	0.75
B7	300×550	4ø20	0.80
B8	300×550	4ø20	0.85
B9	300×550	4ø20	0.90

3.2 Column schedule

Mark	Size	Bars	Ratio
C1	400×400	8ø20	0.60
C2	400×400	8ø20	0.67
C3	400×400	8ø20	0.74
C4	400×400	8ø20	0.81

Mark	Size	Bars	Ratio
C5	400×400	8ø20	0.88
C6	400×400	8ø20	0.95

4 · WORKED SOLUTIONS

4.1 Beams

B12 — 300×550

Level 2 · grid A-2 to A-3
4ø20 · T bf=900 · 300×550
Mu 182 kN·m · Vu 240 kN
Level 2, grid A

01 Flexural demand

Factored moment from the governing envelope.

Mu = (wu L²)/8 = 182.4 kN·m

02 Shear capacity

φ Vn = 0.75(Vc + Vs) = 214.7 kN

Demand exceeds capacity at the support face.

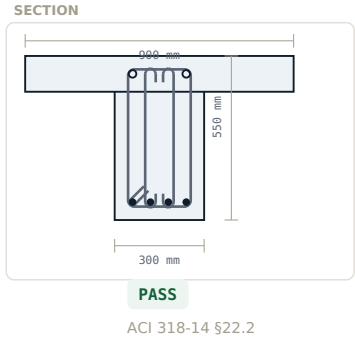
FAIL

stirrups at 150 mm

03 Serviceability

ΔLT = λΔ Δsus + ΔL

ACI 318-14 §24.2



B13 — 300×550

Level 2 · grid A-3 to A-4
3ø20 · 300×550

01 Flexural demand

Factored moment from the governing envelope.

Mu = (wu L²)/8 = 182.4 kN·m

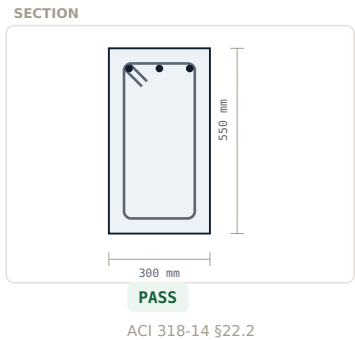
02 Shear capacity

φ Vn = 0.75(Vc + Vs) = 214.7 kN

Demand exceeds capacity at the support face.

FAIL

stirrups at 150 mm



4.2 Columns

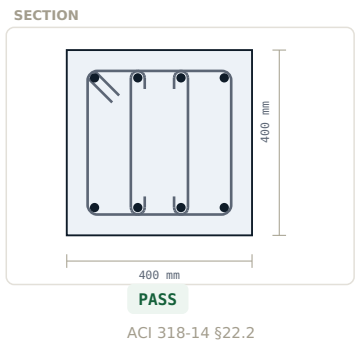
C3 — 400×400

Base to level 1
8ø20 · 400×400
Pu 1840 kN

01 Flexural demand

Factored moment from the governing envelope.

Mu = (wu L²)/8 = 182.4 kN·m



02	Shear capacity	FAIL
	$\phi V_n = 0.75(V_c + V_s) = 214.7 \text{ kN}$ Demand exceeds capacity at the support face.	stirrups at 150 mm
03	Serviceability	ACI 318-14 §24.2
	$\Delta L_T = \lambda \Delta \Delta_{sus} + \Delta L$	